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Bacteroides ovatus alleviates dysbiotic microbiota-induced graft-versus-host disease

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Abstract

Acute lower gastrointestinal GVHD (aLGI-GVHD) is a serious complication of allogeneic hematopoietic stem cell transplantation. Although the intestinal microbiota is associated with the incidence of aLGI-GVHD, how the intestinal microbiota impacts treatment responses in aLGI-GVHD has not been thoroughly studied. In a cohort of patients with aLGI-GVHD (n = 37), we found that non-response to standard therapy with corticosteroids was associated with prior treatment with carbapenem antibiotics and a disrupted fecal microbiome characterized by reduced abundances of *Bacteroides ovatus*. In a murine GVHD model aggravated by carbapenem antibiotics, introducing *B. ovatus* reduced GVHD severity and improved survival. These beneficial effects of *Bacteroides ovatus* were linked to its ability to metabolize dietary polysaccharides into monosaccharides, which suppressed the mucus-degrading capabilities of colonic mucus degraders such as *Bacteroides thetaiotaomicron* and *Akkermansia muciniphila*, thus reducing GVHD-related mortality. Collectively, these findings reveal the importance of microbiota in aLGI-GVHD and therapeutic potential of *B. ovatus*.

Graphical Abstract

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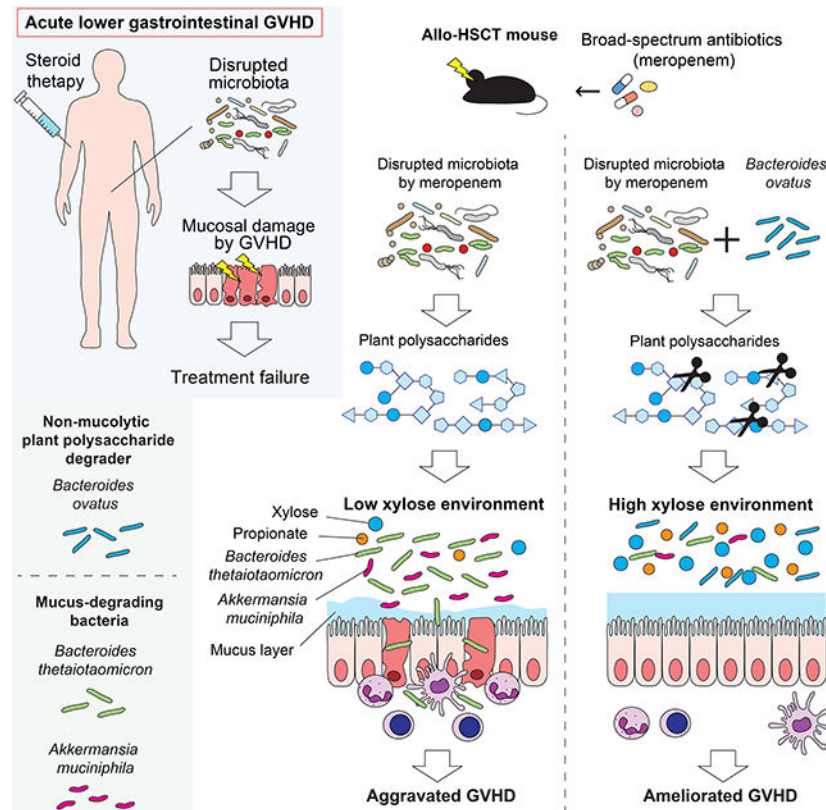
Author contributions

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Declaration of interests

R.R.J. has served as a consultant or advisory board member for Postbiotics Plus, Merck, Microbiome DX, Karius, MaaT Pharma, LIS Cure, Seres, Kaleido, and Prolacta and has received patent license fee or stock options from Seres, Kaleido and Postbiotics Plus. E.J.S. has served as a consultant or advisory board member for Adaptimmune, Axio, Navan, Fibroblasts and FibroBiologics, NY Blood Center, and Celaid Therapeutics and has received patent license fee from Takeda and Affimed. J.U.P. reports research funding, intellectual property fees, and travel reimbursement from Seres Therapeutics, and consulting fees from DaVolterra, CSL Behring, Crestone Inc, and from MaaT Pharma. J.U.P. serves on an Advisory board of and holds equity in Postbiotics Plus Research. J.U.P. has filed intellectual property applications related to the microbiome (reference numbers #62/843,849, #62/977,908, and #15/756,845). Memorial Sloan Kettering Cancer Center (MSK) has financial interests relative to Seres Therapeutics. E.H., M.A.J., J.L.K., and R.R.J. are inventors on a patent application by The University of Texas MD Anderson Cancer Center supported by results of the current study entitled, "Methods and Compositions for Treating Cancer therapy-induced Neutropenic Fever and/or GVHD."



eTOC Blurp

Hayase et al. discover that *Bacteroides ovatus* reduces the severity of graft-versus-host disease (GVHD), a complication of hematopoietic cell transplantation, by suppressing mucus-degrading gut microbes. These beneficial effects of *B. ovatus* are linked to the metabolism of dietary polysaccharides into monosaccharides and highlight the therapeutic potential of *B. ovatus*.

Keywords

Bacteroides ovatus; *Bacteroides thetaiotaomicron*; *Akkermansia muciniphila*; allogeneic hematopoietic stem cell transplantation; graft-versus-host disease; intestinal microbiome; mucus layer; xylose; polysaccharides; polysaccharide utilization loci

Introduction

Graft-versus-host disease (GVHD) is a common complication in patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT) and occurs when donor T cells recognize a patient's tissues as foreign. The intestine is often targeted, and severe acute lower gastrointestinal GVHD (aLGI-GVHD) tends to have a poorer prognosis because of a lower likelihood of responding to therapy¹. Approximately half of aLGI-GVHD cases do not respond to first-line steroid therapy, leading to a high risk for severe complications and reduced overall survival^{2,3}. Novel immune suppression strategies to treat steroid-refractory GVHD have been established, including Janus kinase 1/2 (JAK1/2) inhibitors,

with demonstrated clinical efficacy, although fewer than half of patients with aLGI-GVHD will have a durable response^{4,5}.

The intestinal microbiota is an important modulator of the host immune system^{6,7} and is integral to the pathophysiology of GVHD⁸. Patients undergoing allo-HSCT are at high risk for perturbations in the intestinal microbiota resulting from many factors, chief amongst them exposure to antibiotics for prevention and treatment of bacterial infections post-transplant. Broad-spectrum antibiotics such as carbapenems have been reported to increase the incidence of aLGI-GVHD⁹⁻¹². Recently, fecal microbiota transplantation has been shown to result in improvement in GVHD in steroid-refractory patients¹³⁻¹⁵, suggesting that the intestinal microbiota can modulate aLGI-GVHD treatment responsiveness. It remains unclear, however, how intestinal microbial composition can modulate treatment response of aLGI-GVHD.

In this study, we aimed to determine if the composition of the intestinal microbiota at the onset of aLGI-GVHD is associated with GVHD response to therapy. Our retrospective analysis of 37 aLGI-GVHD patients found that steroid-refractory patients showed greater dysbiosis and lower abundances of *Bacteroides ovatus* at the onset of aLGI-GVHD than responsive patients. In a preclinical model of GVHD, administration of *B. ovatus* improved survival of mice with a disrupted microbiota caused by meropenem treatment. We also found that to mediate this beneficial effect, *B. ovatus* had to be able to degrade xylose-containing polysaccharides and produce abundant monosaccharides including xylose in the colonic lumen. The ability of the microbiota to degrade polysaccharides, especially xylose-containing polysaccharides, may play a key role in improving the intestinal metabolic environment in allo-HSCT and prevent expansion of mucus-degrading bacteria, leading to favorable outcomes of aLGI-GVHD.

Results

***Bacteroides*-enriched microbiome in allo-HSCT patients was associated with a favorable aLGI-GVHD treatment response**

To investigate the potential impact of the intestinal microbiota on aLGI-GVHD severity and treatment response, we prospectively collected samples from patients at MD Anderson Cancer Center who developed aLGI-GVHD in the setting of allo-HSCT from 2017 to 2019. A total of 37 enrolled patients were diagnosed with aLGI-GVHD (Table S1): 28 with classic aLGI-GVHD and 9 with late-onset aLGI-GVHD, by National Institutes of Health consensus criteria¹⁶. We determined treatment response as previously reported¹⁷. All patients received initial therapy with methylprednisolone or prednisone at 2 mg/kg/day followed by tapering per institutional guidelines.

An examination of the microbiome composition of the stool samples using 16S rRNA gene sequencing revealed that our aLGI-GVHD cohort showed a significantly distinct intestinal microbiome at the onset of aLGI-GVHD from that of healthy volunteers, visualized with principal coordinates analysis (PCoA) and tested using permutational multivariate analysis of variance (PERMANOVA) (Figure S1A). In particular, aLGI-GVHD patients showed significantly higher abundances of the genera *Escherichia* *Shigella* and *Enterococcus* and

reductions in the genera *Prevotella* and *Faecalibacterium* (Figure S1B). These results were consistent with previous reports identifying *Escherichia coli* and the genus *Enterococcus* as bacteria that can potentially aggravate GVHD severity^{18,19}.

We next sought to identify subsets within the cohort of aLGI-GVHD patients based on differences in microbiome composition. Using hierarchical clustering of weighted UniFrac beta diversity measures, we identified 2 distinct groups, with 9 patients in cluster 1 and 28 patients in cluster 2 (Figure 1A, B). We found that biological sex and median day of aLGI-GVHD onset were significantly associated with microbiome clustering (Table S2). Biological sex has been known to be associated with the differences in the intestinal microbiota through various factors such as sex hormones and colonic transit time^{20,21}. Regarding the later median day of aLGI-GVHD in cluster 2, we considered the possibility of longer exposure to medications used during the transplantation process. However, antibiotic exposures were not significantly different between clusters (Figure S1C, D). No additional clinical transplant characteristics were significantly different between clusters 1 and 2 (Table S2). Cluster 1 showed a trend towards less dysbiosis, as measured by weighted UniFrac differences from the microbiome of healthy volunteers ($p = 0.05$; Figure 1C, D). Interestingly, we found that cluster 1 included a significantly higher proportion of steroid-responsive GVHD patients than cluster 2 (Figure 1E, F). Performing differential abundance analysis on clusters 1 and 2, we found that cluster 1 was primarily characterized by increased abundance of the genus *Bacteroides* (Figure 1G). Overall, these findings suggested that the composition of the intestinal microbiome may be associated with treatment response in patients with aLGI-GVHD.

We then investigated whether the composition of the intestinal microbiome at the onset of aLGI-GVHD was different between patients who would later be steroid-responsive or steroid-refractory. Our aLGI-GVHD cohort included 20 patients who responded to steroids and 17 patients who did not. We found that steroid-refractory patients were significantly younger than steroid-responsive patients in this cohort (Table S3), in contrast to results of a prior study²². No clinical transplant characteristics besides age were significantly different between responsive and refractory patients. The time from allo-HSCT until onset of aLGI-GVHD was a median of 31.5 days (range, 14–367 days) in steroid-responsive patients and 42 days (13–257 days) in steroid-refractory patients.

We found that steroid-responsive patients showed significantly higher microbial alpha diversity than steroid-refractory patients, but that this diversity was still lower than that of healthy volunteers (Figure 1H). Using PCoA with PERMANOVA testing, we found that the intestinal microbiome composition was significantly different between steroid-responsive and steroid-refractory patients (Figure 1I) and that steroid-refractory patients showed greater dysbiosis than responsive patients, as measured by their weighted UniFrac differences from the microbiome of healthy volunteers (Figure 1I, J). We evaluated the bacterial taxa that were differentially abundant and found that steroid-refractory patients had reductions in the genera *Bacteroides* and *UBA1819* and higher abundances of the genera *Citrobacter*, *Streptococcus*, *Staphylococcus*, and *Enterobacter* (Figure 1K-M).

Overall, these results suggested that alterations of the composition of the intestinal microbiome at clinical presentation of aLGI-GVHD were associated with poor response to therapy, and reductions or absence of *Bacteroides* may contribute to increased aLGI-GVHD severity and treatment failure.

Higher abundance of *Bacteroides ovatus* was significantly associated with favorable aLGI-GVHD treatment outcomes

Allo-HSCT patients are often treated with broad-spectrum antibiotics for febrile neutropenia and other infections that arise before as well as after hematopoietic engraftment. These antibiotics, however, can cause bystander damage to intestinal commensals that are critical for maintaining intestinal homeostasis. Indeed, exposure to broad-spectrum antibiotics such as carbapenems has been linked to an increased incidence of aLGI-GVHD⁹⁻¹². We examined patient antibiotic treatment histories during the period from allo-HSCT to the onset of aLGI-GVHD and looked for associations between treatment with various antibiotic classes and steroid response for GVHD (Figure 2A and Data S1). In our institution, quinolones, cephalosporines, carbapenems and intravenous vancomycin were frequently used in our cohort (Figure 2B). Univariate logistic analysis of frequently-used antibiotics demonstrated that carbapenem exposure was trending towards being associated with steroid-refractory GVHD ($p = 0.05$; Figure 2C). These results are consistent with the hypothesis that antibiotic-mediated microbiome disruption could lead to increased severity of aLGI-GVHD and reduced response rates of GVHD to therapy.

To identify specific species of *Bacteroides* potentially associated with steroid response for aLGI-GVHD, we performed whole-genome sequencing on the subset of fecal samples with sufficient genomic DNA or stool remaining for assessment. In samples from 23 patients, including 11 steroid-responsive patients and 12 steroid-refractory patients, we found that abundances of *B. ovatus* were significantly increased in steroid-responsive patients (Figure 2D). Evaluation of genetic pathways from *Bacteroides* demonstrated that multiple genetic pathways of *Bacteroides* were significantly enriched in steroid-responsive patients but none in steroid-refractory patients (Figure 2E). Interestingly, the top 50 pathways with significantly increased abundances in steroid-responsive patients, including pathways related to amino acid degradation and carbohydrate biosynthesis/degradation, belonged to *B. ovatus* (Figure 2E, F), indicating that *B. ovatus* is particularly associated with steroid-responsive GVHD in patients. To validate these results, we additionally investigated the publicly available whole genome sequencing data of 32 fecal samples collected at the onset of aLGI-GVHD in another institution, Memorial Sloan Kettering Cancer Center²³. This cohort included 26 steroid-responsive GVHD and 6 steroid-refractory GVHD patients (Table S4). Understanding that institutional antibiotic treatment strategies were quite different between the discovery and validation cohorts, we observed that there were no associations between antibiotic exposure and steroid-refractory GVHD in the validation cohort (Figure S1E-F). Microbiome analysis showed significantly higher abundances of the genus *Bacteroides* and the species *B. ovatus* in steroid-responsive patients compared to refractory patients (Figure S1G, H). Evaluating differentially abundant microbial species between steroid-responsive and refractory patients that were observed in our discovery and validation cohorts, we found

that only *B. ovatus* and *Lactococcus lactis* were found in both institutions and both species were significantly associated with steroid-responsive GVHD (Figure S1I).

In summary, carbapenem-mediated microbiome disruption during allo-HSCT might be associated with severe aLGI-GVHD and reduced steroid response for aLGI-GVHD. Our results of 16S rRNA and whole-genome sequencing of patient fecal samples at the onset of aLGI-GVHD implicated a potential beneficial effect of *B. ovatus*, which we further examined in a murine GVHD model.

***B. ovatus* suppressed meropenem-aggravated colonic GVHD in a murine GVHD model**

To investigate whether *B. ovatus* influences GVHD outcomes in a murine GVHD model, we isolated *B. ovatus* from the stool of a healthy volunteer and named the strain MDA-HVS BO001. We assembled the complete genome of MDA-HVS BO001 and confirmed that it was a strain of *B. ovatus*, with 99.4% of the genomic identity of the ATCC strain of *B. ovatus* (ATCC 8483) (Figure S2A). To quantify the genetic similarity between *B. ovatus* from aLGI-GVHD patients to MDA-HVS BO001 and the ATCC strain of *B. ovatus* (ATCC 8483), we built metagenome-assembled genomes (MAGs) from sequencing results of 3 responder patient samples with high abundances of *B. ovatus* (Figure S2B). We then built a database that included 494 *B. ovatus* genomes and assemblies available on GenBank after excluding those tagged as MAGs. We quantified the average nucleotide identity (ANI) between genome pairs, calculated both low and high dimensional embeddings, and visualized the distances by Uniform Manifold Approximation and Projection (UMAP, Figure S2B). MDA-HVS BO001 was classified into cluster E whereas the MAGs of 2 responder patients and the ATCC strain of *B. ovatus* (ATCC 8483) were classified into cluster B and the other responder patient-derived MAG into cluster C (Figure S2B). The mean average distance between assemblies within cluster E was 2.0%, which was large compared to other clusters, indicating that cluster E was a wide cluster (Figure S2C). We found that the distance between MDA-HVS BO001 and each of the MAGs were 2.5 to 2.6% (Figure S3D).

Hereafter, we refer to our isolated *B. ovatus*, MDA-HVS BO001, as *B. ovatus*. Because carbapenem-mediated microbiome disruption prior to aLGI-GVHD onset could be an important determinant of severe GVHD and a potential risk for the development of steroid-refractory GVHD in allo-HSCT patients (Figure 2C), we used a previously-described meropenem-aggravated GVHD murine model²⁴ to evaluate the impact of *B. ovatus* on GVHD severity. Briefly, lethally-irradiated B6D2F1 (H-2^{b/d}) mice were intravenously injected with 5×10^6 bone marrow cells and 5×10^6 splenocytes from major histocompatibility complex (MHC)-mismatched B6 (H-2^b) mice on day 0. Meropenem was administered to the allo-HSCT recipient mice in their drinking water on days 3 to 15 relative to allo-HSCT (Figure 3A). We previously showed that allo-HSCT mice treated with meropenem demonstrated aggravated colonic GVHD in association with loss of the class Clostridia and expansion of *B. theta* compared to mice not treated with meropenem²⁴. *B. theta* is a species of mucus-degrading bacteria that commonly colonizes the intestinal tract of both mice and humans²⁵. In this model, expansion of *B. theta* induces thinning of

the colonic mucus layer and increases bacterial translocation, leading to aggravated colonic GVHD.

To compare mucus-degrading functionality between *B. ovatus* and *B. theta*, we quantified degradation of mucin-derived carbohydrates in vitro using a periodic acid-Schiff (PAS)-based colorimetric assay (Figure S2E). As expected, *B. theta* displayed degradation of mucin-derived carbohydrates, whereas *B. ovatus* did not (Figure S2F), suggesting that *B. ovatus* has less potential to induce mucus-degrading bacteria-related aggravated GVHD, consistent with our prior study²⁶.

Next, to study the effects of *B. ovatus* on GVHD severity, we orally inoculated 2×10^7 colony-forming units of *B. ovatus* into meropenem-treated allo-HSCT recipient mice daily from days 16 to 18 and monitored GVHD severity and survival (Figure 3A). Interestingly, we found that meropenem-treated mice that received *B. ovatus* showed significantly improved survival (Figure 3B). However, the favorable effects of *B. ovatus* were not seen in syngeneic HSCT mice treated with meropenem nor in allo-HSCT mice not treated with meropenem. Meropenem treatment itself in syngeneic-HSCT mice was minimally toxic, with no mortality and only a temporary delay in body weight recovery following irradiation (Figure S3A-E). These data suggested that *B. ovatus* can mitigate the severity of aLGI-GVHD, but only in the context of a meropenem-disrupted microbiota. These findings, together with the finding that expanded *B. theta* after meropenem treatment was associated with aggravated colonic GVHD, indicated that different *Bacteroides* species, which are quite heterogeneous in their metabolic capabilities, can mediate distinct or even opposing effects on aLGI-GVHD^{27,28}. We hypothesized that *B. ovatus* may mitigate GVHD severity via its metabolic capabilities leading to improved intestinal homeostasis, which was supported by our finding in aLGI-GVHD patients that *B. ovatus*-derived pathways were significantly associated with steroid response (Figure 2E).

To elucidate potential mechanisms by which *B. ovatus* mitigated meropenem-aggravated colonic GVHD, we began by examining the microbiota density and composition in both meropenem-treated and untreated allo-HSCT mice with or without introduction of *B. ovatus*. Introduction of *B. ovatus* did not alter bacterial density quantified by 16S rRNA gene quantitative polymerase chain reaction (qPCR), or alpha diversity quantified using the Shannon index, in stool collected on day 21 in meropenem-treated mice or in control allo-HSCT mice untreated with meropenem (Figure 3C, D and Figure S3F, G). The composition of the intestinal microbiome, however, was significantly altered by administration of *B. ovatus* in meropenem-treated mice (Figure 3E). Evaluating for differentially abundant bacterial taxa showed reductions in mucus-degrading bacteria such as *B. theta* and *Akkermansia muciniphila* in meropenem-treated mice that received *B. ovatus* in (Figure 3F-I). We also found that colonization by *B. ovatus* was maintained through day 28 after allo-HSCT (Figure 3J). Consistent with these results, the thickness of the colonic mucus layer was significantly increased in meropenem-treated mice that received *B. ovatus* compared to those without *B. ovatus* (Figure 3K, L). On the other hand, meropenem-untreated allo-HSCT mice showed no significant effects of administration of *B. ovatus* on microbiome composition as a whole, nor on abundances of both *B. theta* and *A. muciniphila* (Figure S3H-L) or on colonic mucus layer thickness (Figure S3M, N). To

consider the possibility that the improvement in the distal colonic mucus layer in the setting of *B. ovatus* introduction was potentially due to increased mucus production, we investigated goblet cell morphology and numbers. We found in meropenem-treated allo-HSCT mice that these were not appreciably changed by *B. ovatus* in the proximal colon, which has recently been demonstrated to be a primary source of mucus in the distal colon²⁹. We did find in the distal colon increased numbers of goblet cells in *B. ovatus*-treated mice (Figure 3K, M, N). Histological scores of the colon quantified by a blinded pathologist were not significantly affected by administration of *B. ovatus*, whereas bacterial translocation into mesenteric lymph nodes (MLNs) was significantly suppressed in those who received *B. ovatus* (Figure 3O, P). These data suggested that *B. ovatus* may ameliorate the colonic mucus layer by suppressing the expansion of mucus-degrading bacteria in mice under certain conditions, such as following meropenem treatment, and also by accelerating the recovery of goblet cells in the distal colon, without clearly impacting on other epithelial cells. Importantly, improvements in colonic mucus after administration of *B. ovatus* was associated with reduced bacterial translocation, which may have contributed to improved survival.

Introducing *B. ovatus* suppressed mucus-degrading functionalities by *B. theta* and *A. muciniphila* in a murine GVHD model

On the basis of our previous finding that meropenem treatment led to changes in carbohydrate concentrations and mucus-degrading functionalities of *B. theta* in our murine GVHD model²⁴, we hypothesized that *B. ovatus* introduction could be impacting the carbohydrate environment and *B. theta* gene expression. We began with investigating the effects of *B. ovatus* on *B. theta* gene expression in meropenem-treated allo-HSCT mice, by performing microbial RNA sequencing of stool samples. We examined RNA reads from *B. theta* and annotated these using the polysaccharide utilization loci (PUL) DataBase³⁰. We found that administration of *B. ovatus* to meropenem-treated allo-HSCT mice led to downregulation in *B. theta* of many PUL genes predicted to contribute to degradation of mucin O-glycans (Figure 4A and Data S1). In contrast, in meropenem-untreated mice, administration of *B. ovatus* did not result in downregulation of any of these PUL genes by *B. theta*, which generally displayed very few transcriptomic changes (Figure S4A and Data S1), supporting our prior finding that *B. theta* in the absence of meropenem treatment did not upregulate mucus-degrading functionalities²⁴.

Since the abundances of *A. muciniphila* were also suppressed by administration of *B. ovatus* in meropenem-treated mice (Figure 3G-I), we next investigated the effect of *B. ovatus* on *A. muciniphila* gene expression in meropenem-treated allo-HSCT mice. Microbial RNA sequencing data showed that administration of *B. ovatus* also led to downregulation of *A. muciniphila* genes predicted to contribute to degradation of mucin O-glycans in meropenem-treated mice (Figure 4B). These results suggested that *B. ovatus* not only suppressed expansion of mucus-degrading bacteria, but also produced downregulation of mucus-degrading functionalities in *B. theta* and *A. muciniphila* in meropenem-treated allo-HSCT mice.

In our previous studies, we found that mucus-degrading functionalities of *B. theta* and *A. muciniphila* are repressed by higher concentrations of ambient molecules, including especially xylose, and propionate, respectively^{24,31}. We thus quantified effects of *B. ovatus* on colonic luminal concentrations of xylose and propionate, as well as other monosaccharides and short-chain fatty acids (SCFAs), using ion chromatography-mass spectrometry (IC-MS). Interestingly, most monosaccharides as well as propionate were increased in meropenem-treated mice that received introduction of *B. ovatus* (Figure 4C-D), indicating that *B. ovatus* may be raising concentrations of monosaccharides and propionate by helping to degrade dietary-derived polysaccharides. To evaluate if *B. ovatus* was sufficient to elevate monosaccharide concentrations by itself without contributions from other intestinal bacteria, we utilized gnotobiotic mouse models. We measured carbohydrate concentrations of colonic luminal contents collected from previously germ-free (GF) mice two weeks after introduction of *B. ovatus*. We found increased concentrations of many monosaccharides in the colonic lumen of mice monocolonized with *B. ovatus*, while GF mice had very low concentrations of nearly all monosaccharides except ribose (Figure S4B). As expected, monosaccharide concentrations in the colonic lumen of meropenem-untreated mice were not significantly affected by *B. ovatus* introduction (Figure S4C). These results suggested that *B. ovatus* functions in the setting of an injured microbiota to elevate concentrations of monosaccharides in the colonic lumen. It has also been reported that *B. ovatus* can produce indole-3-acetic acid which in turn promotes interleukin-22 production from immune cells, leading to decreased colonic inflammation in a murine inflammatory bowel disease model³². We did not observe, however, significant changes in concentrations of tryptophan metabolites due to *B. ovatus* in our model (Figure S4D). Thus, our results indicated that *B. ovatus* is effective in elevating concentrations of monosaccharides and propionate in the colonic lumen of mice compared to mice with an absent or injured microbiota.

Degradation of xylose-containing polysaccharides by *B. ovatus* suppressed mucus-degrading functionalities in *B. theta*

Interestingly, in contrast to *B. theta*, *B. ovatus* is known to have the ability to degrade xylose-containing polysaccharides^{33,34} and we previously found that supplementation of xylose ameliorates GVHD severity in meropenem-treated allogeneic mice by suppressing mucus-degrading functionalities in *B. theta*²⁴. This led us to hypothesize that the ability of *B. ovatus* to degrade xylose-containing polysaccharides could ameliorate GVHD via production of xylose in the colonic lumen. We quantified gene expression of *B. ovatus* using microbial RNA sequencing of stool samples and found that *B. ovatus* in meropenem-untreated mice showed higher expression of PUL genes predicted to perform degradation of xylose-containing polysaccharides (Figure S5A and Data S1), presumably reflecting enriched xylose-containing polysaccharides in the intestinal environment of meropenem-untreated mice compared to meropenem-treated mice (Figure 4C and Figure S4C). These data suggested that expression of *B. ovatus* PUL genes was altered by ambient concentrations of carbohydrates, which are possibly modulated by nutritional intake and metabolism by other bacteria. Despite lower expression of *B. ovatus* PUL genes that degrade xylose-containing polysaccharides in meropenem-treated mice compared to meropenem-untreated mice, meropenem-treated allo-HSCT mice nevertheless showed

reduced abundances of *B. theta*. This could be explained by either sufficient expression of enzymes that degrade xylose-containing polysaccharides or due to higher abundances of *B. ovatus* in meropenem-treated mice. To investigate for potential interactions between *B. ovatus* and *B. theta*, we performed network analysis between PUL genes expressed by *B. ovatus* and *B. theta*. Given our hypothesis that *B. ovatus* was performing metabolic functions that inhibited utilization of mucins by *B. theta*, we were particularly interested in PUL genes of *B. ovatus* that were negatively associated with PUL genes of *B. theta* that participate in degradation of mucin O-glycans. We found that multiple *B. ovatus* genes were negatively correlated with genes belonging to PULs of *B. theta* involved in degradation of mucin O-glycans, including those belonging to PULs involved in metabolizing xylose-containing polysaccharides such as xyloglucan, wheat arabinoxylan, oat spelt xylan, and complex xylans^{33,35} (Figure S5C). These *B. ovatus* genes encoded enzymes predicted to function as beta-glucosidases, beta-galactosidases, and beta-xylosidases (Data S1). This led us to ask if degradation of xylose-containing polysaccharides by *B. ovatus* could produce metabolic byproducts that suppress mucin glycan utilization by *B. theta* in vitro. We evaluated the effects of combining minimal media supplemented with porcine gastric mucin with media conditioned by *B. ovatus* for 48 hours in the presence of wheat arabinoxylan or tamarind xyloglucan, which are composed of xylose, and compared these to wheat starch, which is not composed of xylose (Figure 5A). To measure remaining porcine gastric mucin from culture media, we first removed *B. ovatus*-derived monosaccharides and remaining polysaccharides with ethanol purification, followed by PAS-based colorimetric quantification of mucin post-precipitation (Figure 5B). Interestingly, culture media supplemented with wheat arabinoxylan or tamarind xyloglucan followed by *B. ovatus*-conditioning each significantly suppressed mucin degradation by *B. theta*, while culture medium supplemented with wheat starch followed by *B. ovatus*-conditioning did not suppress mucin degradation by *B. theta* (Figure 5B). These mucolytic inhibitory effects were correlated with concentrations of xylose in culture media after conditioning by *B. ovatus* (Figure 5C). Interestingly, mucin degradation by *A. muciniphila* was suppressed only by culture medium supplemented with tamarind xyloglucan after *B. ovatus*-conditioning, suggesting that suppression of mucus-degrading functionalities in *A. muciniphila* is independent of production of xylose (Figure S5D, E).

We then asked what direct effects *B. ovatus* had in vivo on modulating gene expression in *B. theta*. We turned to gnotobiotic mice and evaluated fecal RNA transcripts in germ-free mice 2 weeks after introducing either *B. theta* alone or *B. ovatus* as well as *B. theta* (Figure 5D). We found that introduction of *B. ovatus* resulted in *B. theta* significantly downregulating PUL genes involved in degradation of mucin O-glycans (Figure 5E). Furthermore, to investigate whether the capability to degrade xylose-containing polysaccharides of *B. ovatus* was necessary to mitigate GVHD severity, we generated a xylan-PUL-deficient strain of *B. ovatus*³³. GF mice that were administered xylan-PUL-deficient *B. ovatus* showed significantly reduced concentrations of xylose in the colonic lumen (Figure 5F, G). We then administered xylan-PUL-deficient *B. ovatus* to meropenem-treated allogeneic mice and evaluated survival (Figure 5H). Interestingly, xylan-PUL-deficient *B. ovatus* failed to improve survival in meropenem-treated mice, while xylan-PUL-wild-type *B. ovatus* (ATCC8483 with genetic deletion of thymidine kinase) significantly improved

GVHD survival in a manner similar to MDA-HVS BO001 in prior experiments (Figure 5I). Altogether, these data suggested that after introduction to meropenem-treated allo-HSCT mice, *B. ovatus* produces a carbohydrate-enriched intestinal environment in the colonic lumen by degrading dietary-derived polysaccharides such as xylose-containing polysaccharides, leading to inhibition of mucin utilization by mucus-degrading bacteria such as *B. theta* and *A. muciniphila*, ultimately resulting in amelioration of disrupted microbiota-induced severe GVHD.

Finally, to genetically assess whether aLGI-GVHD patient-derived *B. ovatus* strains would be predicted to be capable of degrading xylose-containing polysaccharides, similar to MDA-HVS BO001 and the ATCC strain of *B. ovatus* (ATCC 8483), we investigated *B. ovatus* MAGs from 3 steroid-responsive patients. Importantly, we found that MAGs from each patient included genes involved in degradation of xylose-containing polysaccharides, indicating that each patient harbored *B. ovatus* strains with the potential to mitigate aLGI-GVHD through their capability to degrade xylose-containing polysaccharides.

Discussion

Allo-HSCT is a curative therapy for high-risk hematological malignancies, but complications such as infections and GVHD continue to limit its success. The intestinal microbiota is an important modulator of GVHD, and broad-spectrum antibiotics are known to increase the incidence of aLGI-GVHD by compromising several functions of an intact intestinal microbiota, resulting in alterations to the intestinal environment including reduced concentrations of metabolic products in the colonic lumen²⁴. Indeed, the colonic luminal concentrations of SCFAs, especially butyrate and propionate, have been shown to suppress colonic GVHD^{36,37}. The poor prognosis of severe aLGI-GVHD underlines the need to better understand how intestinal microbes can help suppress GVHD in allo-HSCT.

In this study, we investigated the impact of the intestinal microbiota on treatment responsiveness of aLGI-GVHD using clinical microbiome data. In our analysis of prospectively collected fecal samples from a cohort of aLGI-GVHD patients, we found that an altered microbiome profile at presentation of aLGI-GVHD with lower microbial diversity and disrupted composition, accompanied by prior treatment with antibiotics, was significantly associated with developing steroid-refractory GVHD. In contrast, a higher abundance of the commensal species *B. ovatus*, commonly found in normal individuals, was significantly associated with improved GVHD response to steroid therapy. In this study, steroid-responsive patients, who had high abundances of *B. ovatus*, were significantly older at the onset of aLGI-GVHD (Table S3). Since age is one of the known factors that impact the microbiome composition³⁸, the possibility that the older age of steroid-responsive patients was related to the abundances of *B. ovatus* could not be ruled out. However, thus far there are no consistent reports about the relation between the abundance of *Bacteroides* and aging, with some conflicting reports. Notably, we did not see reproducibility of the impact of age on steroid-responsiveness in our validation cohort (Table S4).

In a prior study, *B. ovatus* was associated with a reduced incidence of GVHD³⁹. However, it has not been well-studied whether *B. ovatus* can mechanistically suppress GVHD. Some

prior studies have reported that *B. ovatus* can mediate multiple beneficial functions in maintaining intestinal homeostasis in the host via production of indole-3-acetic acid or sphingolipid production^{32,40}. Here, in a murine model, we found that introduction of *B. ovatus* resulted in improved survival in meropenem-treated allo-HSCT mice but not in meropenem-untreated allo-HSCT mice. This suggested that *B. ovatus* helped suppress GVHD only in hosts with a disrupted microbiota, and that a key function of *B. ovatus* may be related to mechanisms underlying aggravated colonic GVHD in the setting of antibiotic injury. A potential mechanism by which *B. ovatus* improved survival in meropenem-treated allo-HSCT mice was the expression of enzymes by *B. ovatus* that can degrade polysaccharides. These enzymes produce soluble monosaccharides in the colonic lumen that are typically high in concentration in the setting of an intact microbiota and are depleted in antibiotic-treated mice. Unlike *B. ovatus*, *B. theta* is known to be capable of utilizing host-derived glycans^{41,42}, and was found to aggravate colonic GVHD in our prior study²⁴. In this study, we found that in the setting of a meropenem-disrupted microbiota with expansion of mucus-degrading *B. theta*, the introduction of *B. ovatus* improved the thickness of the colonic mucus layer and reduced GVHD-related mortality via polysaccharide degradation, thus producing abundant monosaccharides and improving the intestinal metabolomic environment in allo-HSCT.

A limitation of our study was the relatively small number of patients in our cohort. Because patients vary in terms of when they develop aLGI-GVHD, the timing of stool collection, which was uniformly set at onset of aLGI-GVHD, was variable relative to allo-HSCT. This study did not have time-matched non-GVHD patient samples as controls. As a result, the effects of antibiotic exposures during allo-HSCT and the impact of microbiota disruption due to antibiotics were potentially different for each individual. Also, we found that steroid-refractory patients showed significantly higher histological GVHD grades of the colon than steroid-responsive patients. Changes in the composition of the intestinal microbiota at onset of aLGI-GVHD may be confounded by the severity of GVHD, which can impact on various factors, especially dietary intake, which can have a major impact on intestinal microbiota composition. However, because in this study we did not collect diet information from patients, we are unable to evaluate the impact of interactions between microbiome composition and diet on steroid therapy response for aLGI-GVHD. Importantly, severe mucosal injury from GVHD may itself produce microbiota compositional changes and be associated with a higher likelihood of steroid-resistance.

To better evaluate the potential for causality, we utilized a murine GVHD model combined with in vitro assays and were able to demonstrate that *B. ovatus* ameliorated meropenem-aggravated colonic GVHD via xylose-containing polysaccharide degradation. However, we have not directly demonstrated that the benefit mediated by *B. ovatus* introduction requires intact production of colonic mucin. *B. ovatus* has a broad ability to play a role in not only carbohydrate degradation but also functions in generation of tryptophan metabolites³², sphingolipids⁴⁰, SCFAs⁴³, and bile salt hydrolase⁴⁴ and can impact secretion of fecal immunoglobulin A⁴⁵. In addition, although we have found that *B. ovatus* can ameliorate GVHD aggravated by a dysbiotic microbiota in a murine model, it remains to be seen whether introduction of *B. ovatus* can impact on steroid therapy response for aLGI-GVHD.

Further studies will be needed to fully understand the influence of the intestinal microbiota regarding to response to therapy.

In summary, an antibiotic-disrupted microbiota caused by carbapenems including meropenem increased the severity of intestinal GVHD and was associated with treatment-refractory aLGI-GVHD. Mouse models demonstrated that introducing *B. ovatus* can ameliorate the severity of GVHD in a meropenem-aggravated colonic GVHD model. This understanding of how specific bacteria such as *B. ovatus* can reduce intestinal inflammation should facilitate the development of strategies to better prevent and treat this important complication of allo-HSCT.

STAR Methods

RESOURCE AVAILABILITY

Lead contact—Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Robert R Jenq (rjenq@coh.org).

Materials availability—This study did not generate new unique reagents. All bacterial strains can be obtained from ATCC or as described in the key resources table.

Data and code availability—16S rRNA sequencing data and RNA sequencing data (PRJNA1000552), whole-genome sequencing data of patient fecal samples (PRJNA973955), and complete genome data (PRJNA1022439) have been deposited at Sequence Read Archive (SRA). All data are publicly available as of the date of publication. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Study design—A total of 37 aLGI-GVHD patients who underwent allo-HSCT during 2017 to 2019 at MD Anderson Cancer Center prospectively provided stool samples for our biorepository. Each sample was provided at presentation of symptoms, when an aliquot of the same sample collected to evaluate for infectious causes of diarrhea was stored for later analysis. Acute GVHD was diagnosed by clinical and/or pathological findings and graded according to standard criteria⁴⁶. These patients included 28 with classic aLGI-GVHD and 9 with late-onset aLGI-GVHD by National Institutes of Health consensus criteria¹⁶. We classified patients by steroid responsiveness to GVHD, including 20 patients who were steroid-responsive and 17 patients who were steroid-refractory. We determined treatment response as previously reported¹⁷: briefly, a lack of response on the basis of organ assessment after at least 3 days of high-dose systemic glucocorticoid therapy; a lack of improvement after 7 days; or treatment failure during steroid tapering or an inability to taper the dose to <0.5 mg/kg/day of methylprednisolone. All patients received initial therapy with methylprednisolone or prednisone at 2 mg/kg/day followed by tapering per institutional guidelines. As another cohort, a total of 16 aLGI-GVHD patients who underwent allo-HSCT between 2017 to 2020 at MD Anderson Cancer Center provided stool samples collected on day 14 after allo-HSCT for our biorepository, and these patient stool samples were

analyzed retrospectively. Signed informed consent was provided by all study participants including healthy volunteers, and this study was approved by The University of Texas MD Anderson's Institutional Review Board. As a validation cohort, publicly available whole genome sequencing data from 32 fecal samples collected at the onset of aLGI-GVHD at Memorial Sloan Kettering Cancer Center were investigated²³.

Human samples—Table S1-3 report metadata, including age and sex, corresponding to the 37 aLGI-GVHD patients included in this study. Samples were collected from patients undergoing allo-HSCT and healthy volunteers and stored at 4°C for 24-48 hours until aliquoted for long-term storage at -80°C.

Mice—Female C57BL/6J (B6: H-2^b) and B6D2F1 (H-2^{b/d}, CD45.2⁺) were purchased from The Jackson Laboratory (Bar Harbor, ME). Mice were maintained in specific pathogen-free (SPF) conditions and provided with standard chow (LabDiet 5053) and water. Six- to 12-week-old female C57BL/6 germ-free mice for murine studies were provided by the gnotobiotic facility of Baylor College of Medicine (Houston, TX). Gnotobiotic mice were provided with autoclaved standard chow (LabDiet 5V0F) and water. All animal experiments were performed under the Guide for the Care and Use of Laboratory Animals Published by the National Institutes of Health and was approved by the Institutional Animal Care and Use Committee. Experiments in this manuscript were performed in a non-blinded fashion.

Antibiotics administration—Meropenem was dissolved with phosphate buffer, pH 8.0, and given at a concentration of 0.625 g/L in drinking water from day 3 to day 15 after transplant.

HSCT—Mice received transplants as previously described⁴⁷. In brief, after receiving myeloablative total-body irradiation (11 Gray) delivered in 2 doses at 4-hour intervals, B6D2F1 (H-2^{b/d}) mice were intravenously injected with 5×10^6 bone marrow cells and 5×10^6 splenocytes from allogeneic B6 (H-2^b) donors. Female mice that were 8 to 12 weeks old were allocated randomly to each experimental group, ensuring that the mean body weight was similar across groups. Total body radiotherapy was performed using a Shepherd Mark I, Model 30, ¹³⁷Cs irradiator. Survival after HSCT was monitored daily, and the degree of clinical GVHD was assessed weekly using an established scoring system⁴⁸.

METHOD DETAILS

Histological and immunohistochemistry analysis—For evaluation of mucus thickness, colonic sections containing stool pellets were fixed in methanol-Carnoy fixative composed of methanol (60%), chloroform (30%) and glacial acetic acid (10%) and 5 µm sections were made and stained with periodic acid-Schiff (PAS). Sections were imaged using an Aperio AT2. Mucus thickness of the colonic sections was measured using eSlide Manager Version 12.4.3.5008. Eight measurements per image were taken and averaged over the entire usable colon surface. For pathological analysis, samples of the colon were fixed in 10% formalin, embedded in paraffin, sectioned, and stained with hematoxylin and eosin (H&E). Pathology scores were quantified by a blinded pathologist.

Sequencing of 16S rRNA gene amplicons—Fecal samples that were collected from patients and mice were weighed before DNA isolation. In brief, genomic DNA was isolated using the QIAamp DNA mini kit (51306, Qiagen) according to the manufacturer's protocol, which was modified to include an intensive bead-beating lysis step. The V4 region of the 16S rRNA gene was amplified by PCR from 100 ng of extracted genomic DNA using 515 forward and 806 reverse primer pairs⁴⁹. The quality and quantity of the barcoded amplicons were assessed on an Agilent 4200 TapeStation system and Qubit Fluorometer (Thermo Fisher Scientific), and libraries were prepared after pooling at equimolar ratios. The final libraries were purified using QIAquick gel extraction kit (28706X4, Qiagen) and sequenced with a 2 × 250 base pair paired-end protocol on the Illumina MiSeq platform.

Microbiome data analysis—Sequencing data from paired-end reads were de-multiplexed using QIIME 2⁵⁰. Merging of paired-end reads, dereplicating, and length filtering was performed using VSEARCH2.17.1⁵¹. Following de-noising and chimera calling using the unoise3 command⁵², unique sequences were taxonomically classified with mothur⁵³ using the Silva database⁵⁴ version 138. Weighted UniFrac distances⁵⁵ were determined using QIIME 2, visualized using PCoA, and evaluated for statistical significance using PERMANOVA testing. For differential abundance analysis, abundances of sequences belonging to taxonomical groups were included for analysis using DESeq2 and adjusted for multiple comparisons using the method of Benjamini and Hochberg. Patient microbiome data were classified into 2 clusters using the hcluster function in the amap package in R.

Quantification of fecal bacterial density—Genomic DNA was isolated from stool as described above. qPCR was performed as previously described⁵⁶. In brief, 16S rRNA gene sequences were amplified from total fecal DNA using the primers 926F (5'-AAACTCAAAGGAATTGACGG-3') and 1062R (5'-CTCACRRCACGAGCTGAC-3'). Real-time PCR was carried out in 96-well optical plates on QuantStudio Flex 6 RT-PCR (Thermo Fisher) and KAPA SYBR FAST Master Mix (Roche). The PCR conditions included one initial denaturing step of 10 min at 95°C and 40 cycles of 95°C for 20 sec and 60°C for 1 min. Melting-curve analysis was performed after amplification. To determine bacterial density, a plasmid with a 16S rRNA gene of a murine *Blautia* isolate was generated in the pCR4 backbone and used as a standard.

Culturing of bacteria—*Bacteroides ovatus* (MDA-HVS BO001) was isolated and cultured from a healthy volunteer's stool samples in a Whitley anaerobic chamber (10% H₂, 5% CO₂ and 85% N₂). Human-derived *B. ovatus* (ATCC 8483) and human-derived *B. theta* (ATCC 29148) were purchased from American Type Culture Collection (ATCC) and xylan-PUL deficient *B. ovatus* and wild-type *B. ovatus* (ATCC8483 with gene deletion of thymidine kinase) were provided by Dr. Eric Martens (University of Michigan Medical School, Ann Arbor, Michigan). Mouse-derived *B. theta* (MDA-JAX BT001) and mouse-derived *A. muciniphila* (MDA-JAX AM001) were previously isolated²⁴. Bacterial number was quantified using a Nexcelom Cellometer cell counter with SYTO BC dye and propidium iodide. Bacterial growth experiments were performed in a liquid media, BYEM10, composed of a mix of BHI and M10 supplemented with yeast extract as previously described^{24,31}. Bacteria were cultured up to 24 or 48 hours at a starting

concentration of 1×10^6 bacteria/ml in BYEM10 broth (pH 7.2) with or without 5 mg/ml of porcine gastric mucin (M1778, Sigma-Aldrich), wheat arabinoxylan (wheat flour; low viscosity; Megazyme), xylan (Beechwood; Megazyme), xyloglucan (Tamarind; Megazyme), or starch (wheat; Sigma-Aldrich). Optical densities ($OD_{600\text{ nm}}$) of bacterial cultures were measured with a BioTek Epoch 2 plate reader.

Microbiologic analysis of bacterial translocation—Mesenteric lymph nodes (MLNs) were harvested from mice and homogenized in PBS and cultured anaerobically on BHI plates containing yeast extract and 5% sterilized rumen fluid (Fisher Scientific) for 4 days at 37°C. Colony-forming units (CFUs) were counted and adjusted per organ.

Repairing the thymidine kinase deletion for in vivo colonization—The *B. ovatus* strain used to construct the arabinoxylan and xyloglucan loss of function mutants contained a deletion of the thymidine kinase gene (*tdk*) to facilitate counter selection on 5-fluoro-2-deoxy uridine (FUDR)⁵⁷. To repair this deletion prior to in vivo colonization experiments, we used a cloned version of the *tdk* gene (BT2275) from closely related *B. theta*. A 737 bp fragment containing the intergenic region between the divergently oriented, upstream gene (BT2274), which included the start codon of BT2274, to the stop codon of BT2275 was PCR amplified with primers containing SalI and XbaI restriction sites and cloned into the chromosomal integration vector pNBU2-*ermG*⁵⁸. After sequence confirmation, this construct was introduced by conjugation from *E. coli* S17-1 λ *pir* into *B. ovatus* Δtdk or the arabinoxylan and xyloglucan mutants, which also lack *tdk*, and positive transconjugants were selected on medium containing 25 μ g/ml erythromycin. Successful restoration of thymidine kinase function was measured by plating dilutions (10^0 - 10^{-9}) on BHI-blood plates containing no drug or 200 μ g/ml FUDR to ascertain that susceptibility to this drug was restored by the transgene.

Mucin degradation assay—Concentrations of mucin glycans remaining in culture supernatants were determined by a PAS-based colorimetric assay as previously described^{24,31}. Briefly, culture supernatants were centrifuged at 20,000g for 10 minutes at 4°C and collected. To perform mucin precipitation, 500 μ l of culture supernatants was mixed with 1 ml of molecular grade ethanol and incubated at -30°C for overnight. Culture supernatants were centrifuged at 20,000g for 10 minutes at 4°C. Mucin-containing pellets were washed with 1 ml of molecular grade ethanol twice and resuspended in 500 μ l of PBS. A total of 10 μ l of washed culture supernatants was transferred into a round-bottom 96-well plate containing 15 μ l of PBS. Serially diluted porcine gastric mucin (Sigma-Aldrich) standards were prepared. Freshly prepared 0.06% periodic acid in 7% acetic acid was added and incubated at 37°C for 90 min, followed by 100 μ l of Schiff's reagent (84655, Sigma-Aldrich) and incubation at room temperature for 40 min. Absorbance was measured at 550 nm using a BioTek Synergy HTX plate reader.

Analysis of carbohydrates by IC-MS—To determine the relative concentrations of carbohydrates in mouse fecal samples, extracts were prepared and analyzed by ultrahigh-resolution mass spectrometry. Fecal pellets were homogenized with a Precellys Tissue Homogenizer. Metabolites were extracted using 1 ml of ice-cold 80/20 (v/v) methanol/

water. Extracts were centrifuged at 17,000g for 5 min at 4°C, and supernatants were transferred to clean tubes, followed by evaporation to dryness under nitrogen. Dried extracts were reconstituted in deionized water, and 5 µl was injected for analysis by IC-MS. IC mobile phase A (MPA; weak) was water, and mobile phase B (MPB; strong) was water containing 100 mM KOH. A Thermo Scientific Dionex ICS-5000+ system included a Thermo CarboPac PA20-Fast column (4 µm particle size, 100 × 2 mm) with the column compartment kept at 30°C. The autosampler tray was chilled to 4°C. The mobile phase flow rate was 200 µl/min, and the gradient elution program was: 0-0.5 min, 1% MPB; 0.5-10 min, 1%-5% MPB; 10-15 min, 5%-95% MPB; 15-20 min, 95% MPB; 20.5-25, 95-1% MPB. The total run time was 25 min. To assist the desolvation for better sensitivity, methanol was delivered by an external pump and combined with the eluent via a low dead volume mixing tee. Data were acquired using a Thermo Orbitrap Fusion Tribrid Mass Spectrometer under ESI negative ionization mode at a resolution of 240,000. Raw data files were imported to Thermo TraceFinder and Compound Discoverer software for spectrum database analysis. The relative concentrations of each metabolite were normalized by sample weight.

Analysis of short-chain fatty acids profiling by IC-MS—To determine the relative abundance of short chain fatty acids in mouse feces samples, extracts were prepared and analyzed by ultra-high resolution mass spectrometry (HRMS). Fecal pellets were homogenized with a Precellys Tissue Homogenizer. Metabolites were extracted using 1 mL ice-cold 0.1% Ammonium hydroxide in 80/20 (v/v) methanol/water. Extracts were centrifuged at 17,000 g for 5 min at 4°C, and supernatants were transferred to clean tubes, followed by evaporation to dryness under nitrogen. Dried extracts were reconstituted in deionized water, and 5 µL was injected for analysis by IC-MS. IC mobile phase A (MPA; weak) was water, and mobile phase B (MPB; strong) was water containing 100 mM KOH. A Thermo Scientific Dionex ICS-5000+ system included a Thermo IonPac AS11 column (4 µm particle size, 250 × 2 mm) with column compartment kept at 30°C. The autosampler tray was chilled to 4°C. The mobile phase flow rate was 360 µL/min, and the gradient elution program was: 0-5 min, 1% MPB; 5-25 min, 1-35% MPB; 25-39 min, 35-99% MPB; 39-49 min, 99% MPB; 49-50, 99-1% MPB. The total run time was 50 min. To assist the desolvation for better sensitivity, methanol was delivered by an external pump and combined with the eluent via a low dead volume mixing tee. Data were acquired using a Thermo Orbitrap Fusion Tribrid Mass Spectrometer under ESI negative ionization mode at a resolution of 240,000. Raw data files were imported to Thermo Trace Finder and Compound Discoverer software for spectrum database analysis. The relative abundance of each metabolite was normalized by sample weight.

Analysis of tryptophan metabolites by LC-HRMS—To determine concentrations of tryptophan metabolites in mouse fecal samples, extracts were prepared and analyzed by liquid chromatography coupled with high-resolution mass spectrometry (LC-HRMS). Approximately 50 mg of stool was pulverized on liquid nitrogen, then homogenized with Precellys Tissue Homogenizer. Metabolites were extracted using 0.5 ml of ice-cold 50/50 (v/v) methanol/acetonitrile followed by 0.5 mL 0.1% formic acid in 50/50 (v/v) Acetonitrile/Water. Extracts were centrifuged at 17,000g for 5 min at 4°C, and supernatants were transferred to clean tubes, followed by evaporation to dryness under nitrogen. Samples were

then reconstituted in 50/50 (v/v) methanol/water, then 10 µl was injected into a Thermo Vanquish liquid chromatography (LC) system containing a Waters XSelect HSS T3 2.1 × 150 mm column with 2.5-µm particle size. MPA was 0.1% formic acid in water. MPB was 100% methanol. The flow rate was 200 µl/min (at 35°C), and the gradient conditions were: initial 5% MPB, increased to 95% MPB at 15 min, held at 95% MPB for 5 min, and returned to initial conditions and equilibrated for 5 min. The total run time was 25 min. Data were acquired using a Thermo Orbitrap Fusion Tribrid mass spectrometer under ESI positive and negative ionization modes at a resolution of 240,000 with full scan mode. Raw data files were imported into Thermo TraceFinder software for final analysis. The relative concentration of each compound was normalized by stool weight.

Whole-genome sequencing of patient fecal samples—Genomic DNA was isolated from patient fecal samples and purified using a Qiagen Genomic-tip 20/G column, according to the manufacturer's instructions. For short-read Illumina sequencing, libraries were constructed with a Nextera DNA Flex Library Prep Kit (Illumina), according to the manufacturer's protocol. All libraries were quantified with a TapeStation and pooled in equal molar ratios. The final libraries were sequenced with the NovaSeq 6000 platform (Illumina) to produce 2×150 bp paired-end reads, resulting in ~5 Gb per sample. In sequencing analysis, sequence reads were filtered by their quality using VSEARCH 2.17.1. The abundance of taxa, microbial metabolic pathways, and gene expression was profiled using HUMAnN3. Differential expression profiles were analyzed with DESeq2 in R and P values were corrected using the method of Benjamini and Hochberg.

Whole-genome sequencing of *B. ovatus* (MDA-HVS BO001)—*B. ovatus* (MDA-HVS BO001) genomic DNA was isolated and purified using a Qiagen Genomic-tip 20/G column, according to the manufacturer's instructions. For short-read Illumina sequencing, libraries were constructed with a Nextera DNA Flex Library Prep Kit (Illumina, San Diego, CA, USA), according to the manufacturer's protocol. All libraries were quantified with a TapeStation and pooled in equal molar ratios. The final libraries were sequenced with the NovaSeq 6000 platform (Illumina) to produce 2×150 bp paired-end reads, resulting in ~5 Gb per sample. For long-read Nanopore sequencing, 500 ng of genomic DNA was used for library preparation using the Rapid Sequencing Kit (SQK-RAD004, Oxford Nanopore Technologies). Libraries were loaded into a FLO-MIN106 flow-cell for a 24-h sequencing run on a MinION sequencer platform (Oxford Nanopore Technologies, Oxford, UK). Data acquisition and real-time base calling were carried out by the MinKNOW software version 3.6.5. The fastq files were generated from basecalled sequencing fast5 reads.

Hybrid assembly and genome annotation of *B. ovatus* (MDA-HVS BO001)—To assemble the complete genome of *B. ovatus*, Flye version 2.8.2⁵⁹ was used with long reads (Nanopore) and short reads (NovaSeq) combined using default settings. The similarities of the genome of MDA-HVS BO001 to other reference genomes was calculated using blastn for *B. ovatus* (ATCC 8483)⁶⁰. Open reading frames of *B. ovatus* (MDA-HVS BO001) were identified using prokka⁶¹. The genome of *B. ovatus* and open reading frames were depicted using DNA plotter software⁶².

Metagenome-assembled genomes (MAGs) from patient stool datasets—

To recover *B. ovatus* MAGs from patient's stool metagenomic datasets that were previously determined to have a high abundance of *B. ovatus*, reads were assembled using MEGAHIT⁶³. Resulting contigs were mapped to a databases of 494 non-MAG *B. ovatus* genomes available at GenBank⁶⁴. Contigs mapping >95% to all *B. ovatus* genomes were retained for further consideration. The matching contigs were mapped to all genomes within the Bacteroidales class. Contigs that aligned better to Bacteroidales species other than *B. ovatus* were removed. A final filter using Kraken2⁶⁵ removed database contaminants. To cluster the *B. ovatus* genomes, we used MASH to determine the average nucleotide identity (ANI) between all *B. ovatus* genomes (MAGs + 494 entries). Calculated distances were input into Uniform Manifold Approximation and Projection (UMAP)⁶⁶ and a high and low dimensional embedding were calculated. The high dimensional embedding was used by HDBScan⁶⁷ to compute cluster while the low dimensional embedding was used for plotting.

RNA sequencing and analysis—

Approximately 30 mg of stool was freshly collected in 700 µl of ice-cold QIAzol containing 200 µl of 0.1-mm-diameter Zirconia Silica beads (11079101z, BioSpec). Samples were bead beaten twice for 2 min with a 30-s interval recovery. Samples were then centrifuged at 12,000g for 1 min, and the supernatant was collected for RNA isolation using the RNeasy mini kit (74104, Qiagen). RNA was treated on column with DNase I (79254, Qiagen) to eliminate contaminating genomic DNA. RNA quantity and quality were determined using an Agilent 4200 TapeStation system (Agilent). A total of 250 ng of total RNA from mouse stools was used to construct libraries using the Universal Prokaryotic RNA-Seq Library Preparation Kit (9367-32, Tecan) with Unique Dual Indexes (S02480-FG, Tecan), following the manufacturer's protocol. The cDNA libraries were sequenced on the Illumina NovaSeq 6000 system to produce 2 × 150 bp paired-end reads. Sequence data were demultiplexed using QIIME 2⁵⁰ and their qualities were checked using VSEARCH 2.17.1⁵¹. Data were filtered and truncated by quality with VSEARCH default settings. The total reads of mouse stool samples were 160896223 ± 93489752 (mean ± standard deviation). Sequences of ribosomal RNA were removed using BWA software against prokaryotic ribosomal RNA sequences from prokaryotic RefSeq genomes⁶⁸. Sequences of interest were further identified using diamond software version 0.9.24⁶⁹ to align against PUL genes. Features with percentage identity less than 80% were excluded. The total counts of bacterial isolated samples were 360932 ± 284308 and 966485 ± 617495 in *B. ovatus* and *B. theta*, respectively (mean ± standard deviation). Aligned mRNA expression changes were calculated using the DESeq2 in R software version 4.1.2 via RStudio version 2022.02.0 Build 443. P values < 0.05 after adjustment using the method of Benjamini and Hochberg were considered statistically significant.

Network analysis using bacterial RNA transcripts—

The expressions of PUL genes of *B. theta* and *B. ovatus* in meropenem-untreated and -treated mice that received *B. ovatus* were normalized to control median in each experiment. PUL genes with average expression rates of 1% or less in each group were excluded from the network analysis. Each data set was logit transformed, and then r and p values were calculated by Pearson correlation analysis between *B. theta* and *B. ovatus* PUL genes. P values were adjusted using the method of Benjamini and Hochberg. PUL gene combinations showing a corrected p value of

0.05 or less with a negative r value were depicted using Cytoscape⁷⁰ for *B. theta* PUL genes known to degrade mucin-O-glycan.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis—Data were checked for normality and similar variances between groups, and Student t-tests were used when appropriate. Mann-Whitney U tests were used to compare data between two groups when the data did not follow a normal distribution. Kaplan-Meier curves were used to depict survival probabilities, and the log-rank test was applied to compare survival curves. For clinical data analysis, non-repeated ANOVA was used to compare continuous variables, while chi-square or Fisher exact tests were used to analyze the frequency distribution between categorical variables. Analyses were performed using R software version 4.1.2 and Prism version 9.0 (GraphPad Software). P values < 0.05 were considered statistically significant.

Supplementary Material

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Highlights

- Clinical steroid-refractory GVHD is associated with reduced *Bacteroides ovatus*.
- Introduction of *B. ovatus* after meropenem reduces experimental GVHD-related mortality.
- The beneficial effects of *B. ovatus* are linked to its ability to produce xylose.
- *B. ovatus* suppresses the mucus-degrading capabilities of colonic mucus degraders.

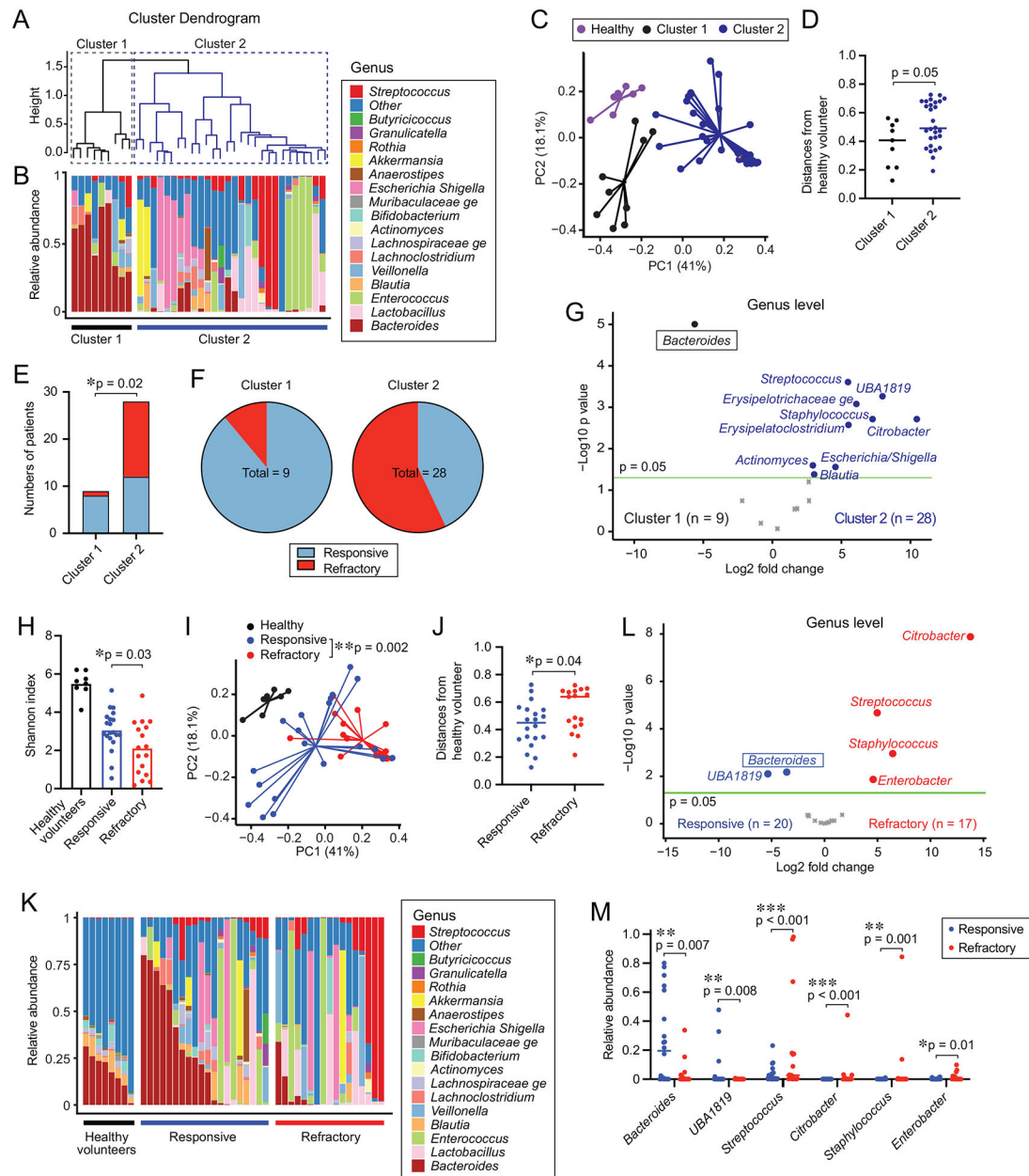


Figure 1. The intestinal microbiome of steroid-refractory aLGI-GVHD patients shows significantly more dysbiosis than that of steroid-responsive aLGI-GVHD patients. (A-D, G-M) The intestinal microbiome analyzed by 16S rRNA sequencing in patient stool samples collected at presentation with aLGI-GVHD. (A) Cluster dendrogram analyzed using hierarchical clustering of weighted UniFrac distances. (B) The microbiome composition shown as stacked bar graphs. (C) PCoA of fecal samples collected from healthy volunteers and two clusters of aLGI-GVHD patients. (D) Distances from healthy volunteers quantified by weighted UniFrac. (E) Numbers of patients with steroid-responsive and -refractory GVHD. (F) Proportions of patients with steroid-responsive and -refractory GVHD. (G) Volcano plot of differentially abundant genera between clusters 1 and 2. (H) Alpha diversity quantified by Shannon index. (I) Principal coordinates analysis (PCoA) of fecal samples

collected from healthy volunteers, steroid-responsive, and steroid-refractory patients. **(J)** Distances from healthy volunteers quantified by weighted UniFrac. **(K)** The composition of the intestinal microbiome. **(L)** Volcano plot of differentially abundant genera. **(M)** Relative abundances of genera that were significantly different between steroid-responsive and -refractory aLGI-GVHD.

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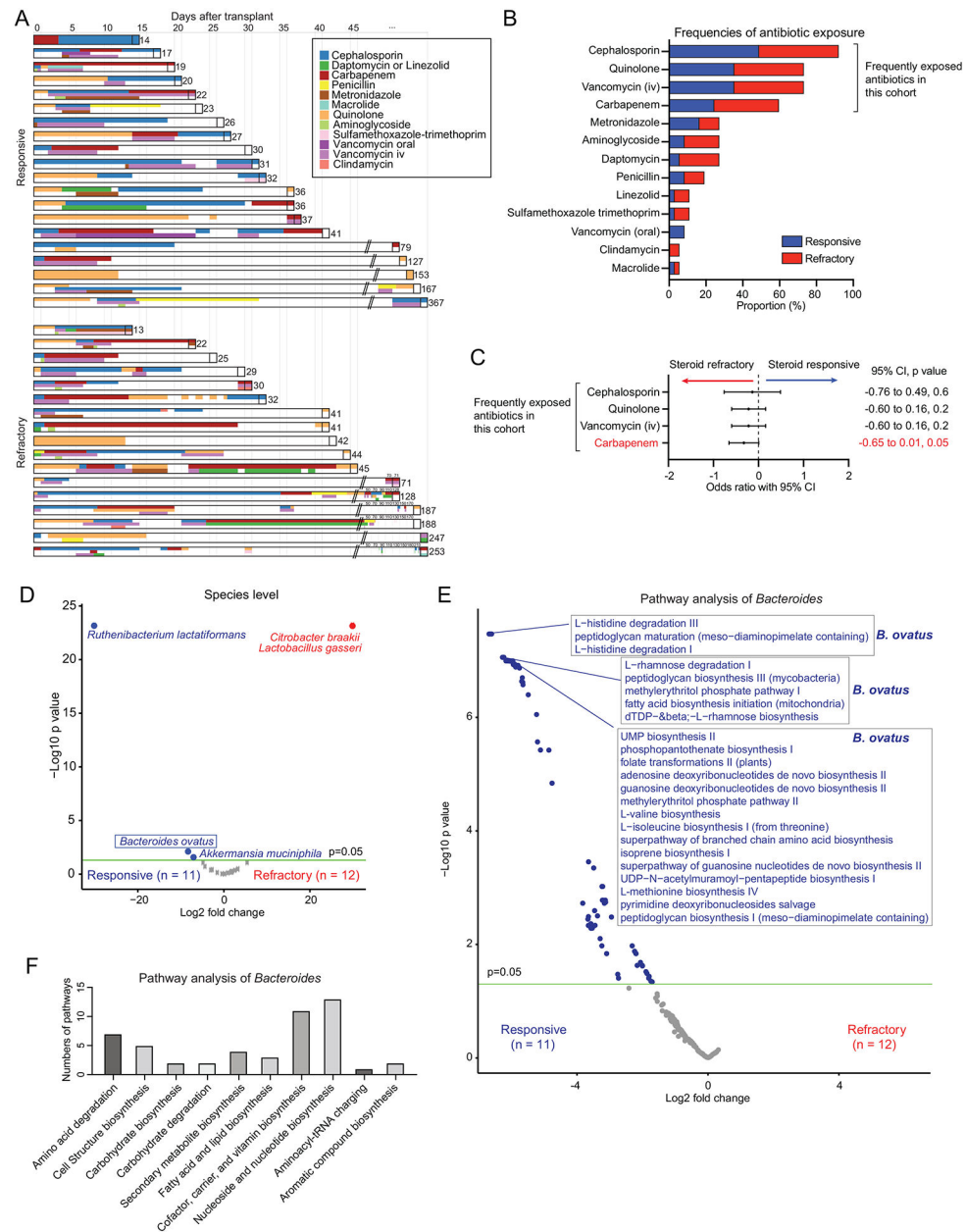


Figure 2. Higher abundances of *Bacteroides ovatus* and *B. ovatus*-derived pathways are associated with steroid-responsive GVHD.

(A) Graphical summary of antibiotics used in individual patients between hematopoietic stem cell transplant (HSCT) and onset of GVHD. (B) Proportions of patients with antibiotic exposures between HSCT and onset of GVHD. (C) Univariate logistic regression analysis for associations between antibiotic exposures and steroid-refractory GVHD. (D-F) Data analyzed by DNA shotgun sequencing of fecal samples collected from aLGI-GVHD patients (steroid-responsive; n=11, steroid-refractory; n=12). (D) Volcano plot of differentially abundant species between steroid-responsive and -refractory GVHD. (E) Volcano plot of differentially abundant pathways of the genus *Bacteroides*. (F) The top 50 subclasses of differentially abundant pathways of the genus *Bacteroides*.

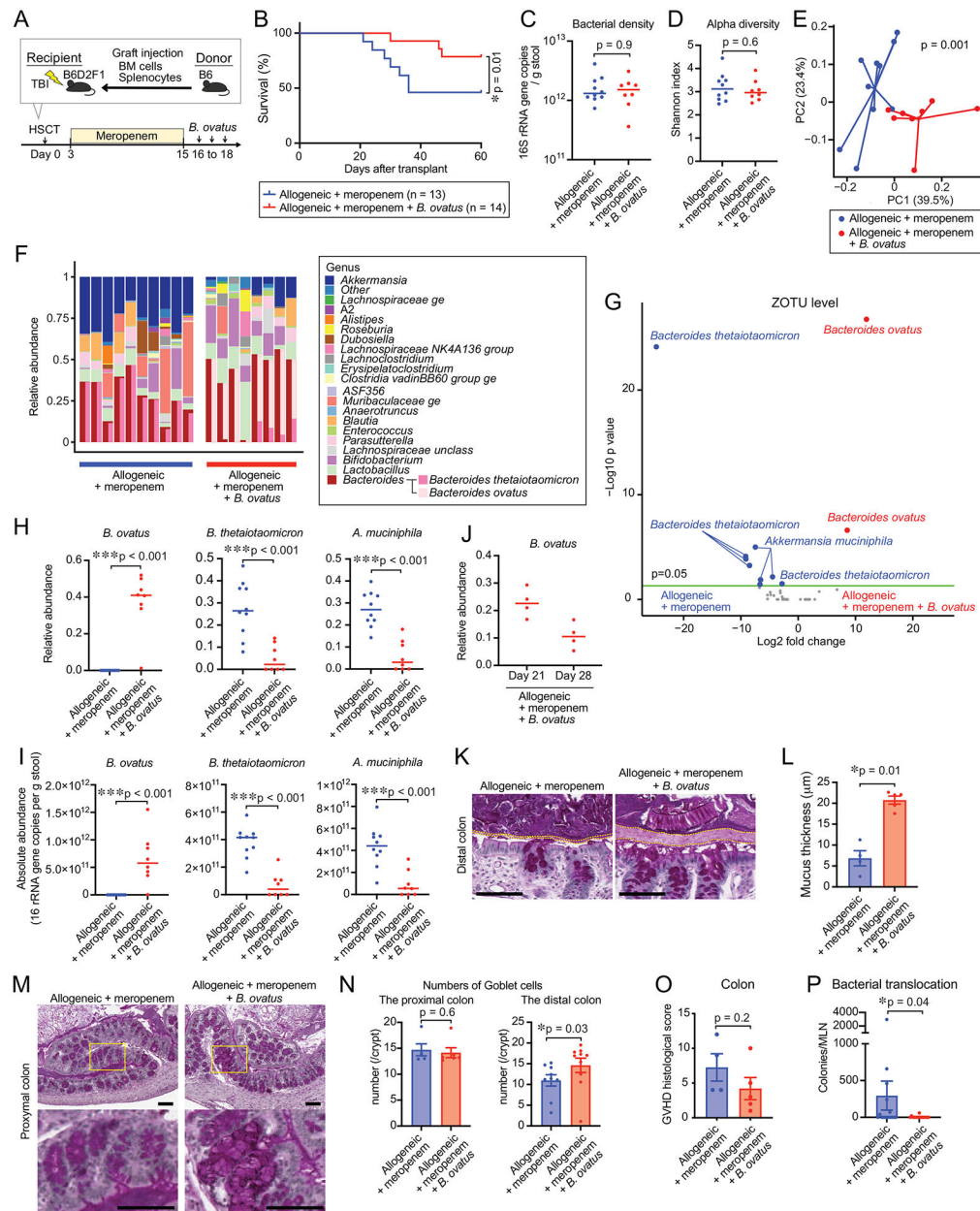


Figure 3. Oral introduction of *Bacteroides ovatus* reduces GVHD-related mortality in mice with meropenem-aggravated colonic GVHD.

(A) Experimental schema of murine GVHD model using meropenem treatment followed by oral gavage of 20 million colony-forming units of *B. ovatus* daily for 3 days. (B) Overall survival after allo-HSCT. Data are combined from two independent experiments. (C) Bacterial densities of mouse stool samples collected on day 21. Bacterial densities were measured by 16S rRNA gene qPCR. (D) Alpha diversity, measured by the Shannon index, was quantified in fecal samples. (E) Principal coordinates analysis (PCoA) of fecal samples. (F) Bacterial genera composition of fecal samples. (G) Volcano plot of differentially abundant zero-radius operational taxonomic units (ZOTUs). (H) Relative abundances of *B. ovatus* (left), *B. theta* (middle) and *A. muciniphila* (right). (I) Absolute abundances of

B. ovatus (left), *B. theta* (middle) and *A. muciniphila* (right). **(J)** Relative abundances of *B. ovatus* in mouse stool samples collected on days 21 and 28. **(K)** Periodic acid-Schiff (PAS) staining of histological distal colon sections collected on day 23. Bar, 100 μm . The areas inside dotted lines indicate the inner dense colonic mucus layer. **(L)** Mucus thickness on day 23. Data are shown from one representative experiment. **(M)** PAS staining of histological proximal colon sections collected on day 21. Areas in the yellow squares are magnified and shown to the bottom of the original images. Bar, 100 μm . **(N)** Numbers of goblet cells per crypt in the proximal or distal colon. **(O)** GVHD histology scores of the colon harvested on day 28. GVHD histology scores were quantified by a blinded pathologist. **(P)** Numbers of bacterial CFUs cultivated from mesenteric lymph nodes (MLNs) on day 21. Combined data from three independent experiments are shown.

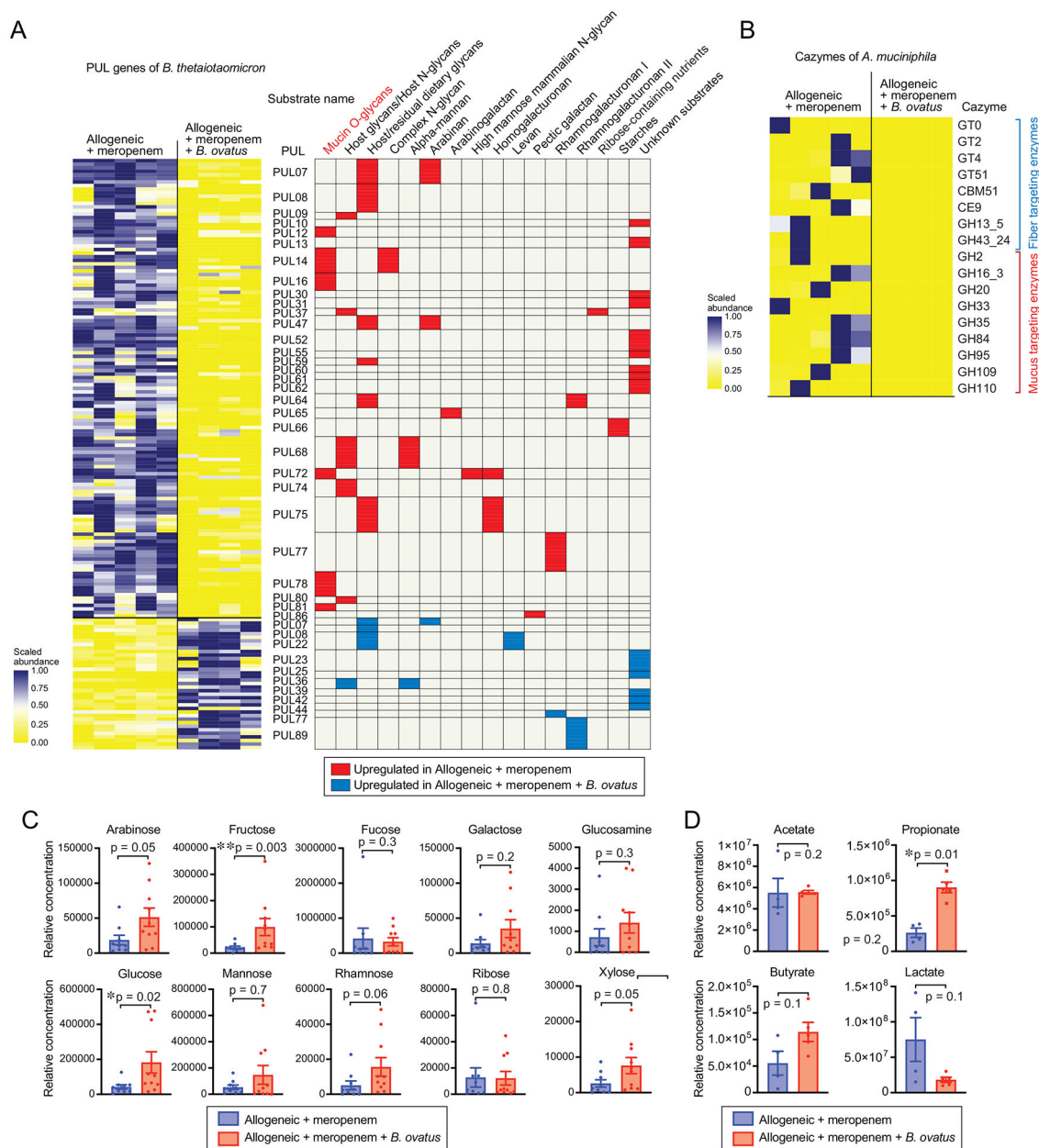


Figure 4. Expression of predicted mucus-degrading enzymes by *Bacteroides theta*tao*micron* and *Akkermansia muciniphila* is suppressed in meropenem-treated mice after administration of *Bacteroides ovatus*.

(A) Heatmap showing scaled relative expression levels of polysaccharide utilization loci (PUL) genes in *B. theta* RNA transcripts sequenced from stool collected from meropenem-treated allo-HSCT mice with or without administration of *B. ovatus* on day 21. Right: Significantly altered PUL genes and their substrates. (B) Heatmap showing scaled relative expression of carbohydrate-active enzymes (CAZymes) by *A. muciniphila* in stool collected from meropenem-treated allo-HSCT mice with or without administration of *B. ovatus* on day 21. (C) Relative concentrations of monosaccharides of supernatants from colonic luminal content collected from meropenem-treated allo-HSCT mice with or without

administration of *B. ovatus* on day 23 measured by ion chromatography-mass spectrometry (IC-MS). Combined data from two independent experiments are shown as means $\pm$ SEM.

(D) Relative abundances of short-chain fatty acids of supernatants from colonic luminal content collected from meropenem-treated allo-HSCT mice with or without administration of *B. ovatus* on day 23 measured by IC-MS.

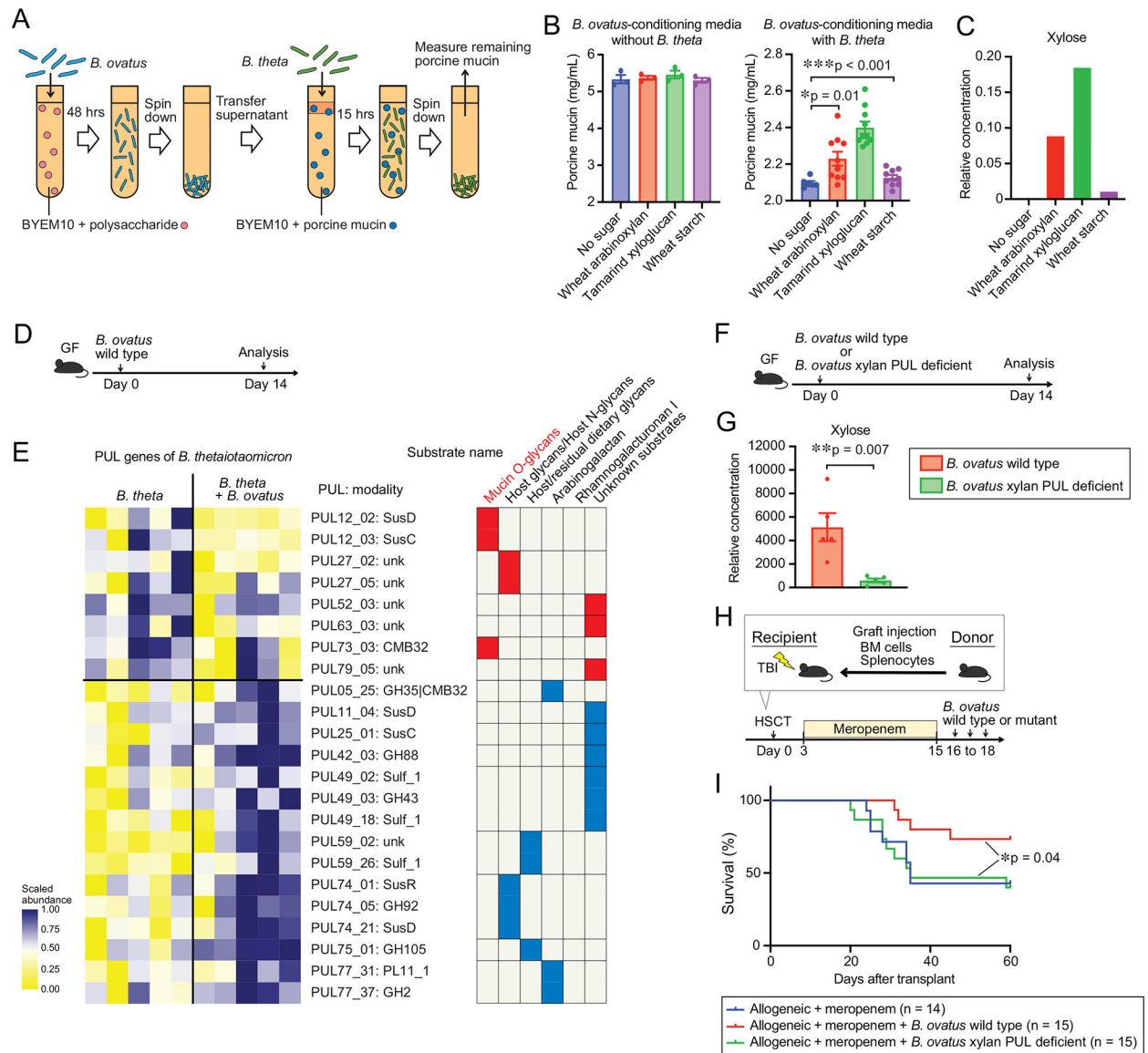


Figure 5. Byproducts of xylose-containing polysaccharides metabolized by *Bacteroides ovatus* suppress mucus-degrading functionality in *Bacteroides thetaiotaomicron*.

(A) Experimental schema of in vitro bacterial culture assay using *B. ovatus* (MDA-HVS BO001) cultured in minimum nutrition medium with each polysaccharide and *B. theta* (MDA-JAX BT001) cultured in BYEM10 with porcine gastric mucin. (B) Concentrations of porcine gastric mucin in the culture supernatant without (left) or with *B. theta* (right) were determined using a PAS-based colorimetric assay. Combined data from two independent experiments are shown as means $\pm$ SEM. (C) Relative concentrations of monosaccharides of the *B. ovatus* culture supernatant with each polysaccharide measured by ion chromatography-mass spectrometry (IC-MS). (D) Experimental schema of gnotobiotic model using introduction of 20 million colony-forming units of *B. ovatus* (MDA-HVS BO001). (E) Heatmap showing scaled relative expression of polysaccharide utilization loci (PUL) genes by *B. theta* from *B. theta* (ATCC 29148)-colonized gnotobiotic mice with or

without co-administration of *B. ovatus*. Expression was evaluated on day 14 after bacterial introduction to germ-free mice. Right: Significantly altered PUL genes and their substrates. **(F)** Experimental schema of gnotobiotic model using introduction of 20 million colony-forming units of wild-type *B. ovatus* (ATCC8483 with gene deletion of thymidine kinase) or xylan-PUL deficient *B. ovatus*. **(G)** Relative concentrations of xylose in supernatants from colonic luminal contents collected from gnotobiotic mice with administration of wild-type *B. ovatus* (ATCC8483 with gene deletion of thymidine kinase) or xylan-PUL deficient *B. ovatus* on day 14 measured by ion chromatography-mass spectrometry (IC-MS). **(H)** Experimental schema of murine GVHD model using meropenem treatment followed by oral gavage of 20 million colony-forming units of wild-type *B. ovatus* (ATCC8483 with gene deletion of thymidine kinase) or xylan-PUL deficient *B. ovatus* daily for 3 days. **(I)** Overall survival after allo-HSCT. Data are combined from three independent experiments.

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Bacterial and virus strains		
<i>Bacteroides ovatus</i> (MDA-HVS BO001)	This study	N/A
<i>Bacteroides ovatus</i> with gene deletion of thymidine kinase	This study	N/A
<i>Bacteroides ovatus</i> xylan-PUL deficient	This study	N/A
<i>Bacteroides thetaiotaomicron</i> (MDA-JAX BT001)	This study	N/A
<i>Bacteroides thetaiotaomicron</i>	ATCC	Cat# ATCC 29148
<i>Akkermansia muciniphila</i> (MDA-JAX AM001)	This study	N/A
Biological samples		
Human participants/stool samples	This study	N/A
Chemicals, peptides, and recombinant proteins		
Food chow (LabDiet PicoLab Rodent Diet 20)	LabDiet	Cat# 5053
Food chow (LabDiet 5V0F)	LabDiet	Cat# 5V0F
Wheat arabinoxylan (wheat flour; low viscosity)	Megazyme	Cat# P-WAXYL
Xyloglucan (Tamarind)	Megazyme	Cat# P-XYGLN
Starch (wheat)	Sigma-Aldrich	Cat# S5127
Porcine gastric mucin	Sigma-Aldrich	Cat# M1778
Sterilized Rumen fluid	Fisher Scientific	Cat# NC1530570
Schiff's reagent for aldehydes	Sigma-Aldrich	Cat# 84655
RNase-Free DNase Set	Qiagen	Cat# 79254
Critical commercial assays		
QIAamp DNA mini kit	Qiagen	Cat# 51306
QIAquick gel extraction kit	Qiagen	Cat# 28706X4
QIAGEN Genomic-tip 20/G	Qiagen	Cat# 10223
KAPA SYBR FAST Master Mix	Roche	Cat# 07959389001
SYTO™ BC Green Fluorescent Nucleic Acid Stain	Thermo Fisher	Cat# S34855
Propidium Iodide Solution	BioLegend	Cat# 421301
Nextera DNA Flex Library Prep Kit	Illumina	Cat# 20018704
Universal Prokaryotic RNA-Seq	Tecan	Cat# 9367-32
UDI 96-Plex Adaptor Plate	Tecan	Cat# S02480-FG
MiSeq Reagent Kit v2 (300-cycles)	Illumina	Cat# MS-102-2002
NovaSeq 6000 SP Reagent Kit v1.5	Illumina	Cat# 20028400
Ovation® Complete Prokaryotic RNA-Seq DR Multiplex System	NuGEN Technologies	Cat# 0326-32/ 0327-32
RNeasy mini kit	Qiagen	Cat# 74104

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Rapid Sequencing Kit	Oxford Nanopore	Cat# SQO-RAD004
High-Capacity cDNA Reverse Transcription Kit	Thermo Fisher	Cat# 4368814
Deposited data		
16S rRNA sequencing data of mouse fecal samples	This study	SRA PRJNA1000552
16S rRNA sequencing data of patient fecal samples	This study	SRA PRJNA1000552
Whole-genome sequencing data of patient fecal samples	This study	SRA PRJNA973955
Microbial RNA sequencing data	This study	SRA PRJNA1000552
Complete genome of MDA-HVS BO001	This study	SRA PRJNA1022439
Experimental models: Cell lines		
Mouse: C57BL/6J	The Jackson Laboratory	JAX: 000664
Mouse: B6D2F1/J	The Jackson Laboratory	JAX: 100006
Experimental models: Organisms/strains		
515 forward and 806 reverse primer pairs	Caporaso et al., 2012	N/A
926F (5'-AAACTCAAAGGAATTGACGG-3')	Yang et al., 2015	N/A
1062R (5'-CTCACRRACAGAGCTGAC-3')	Yang et al., 2015	N/A
Oligonucleotides		
Recombinant DNA		
Software and algorithms		
Flow Jo	FlowJo LLC	RRID:SCR_008520
GraphPad Prism	GraphPad Software	RRID:SCR_002798
Flye version 2.8.2	Kolmogorov et al., 2019	https://github.com/fenderglass/Flye.git
QIIME2	Caporaso et al., 2019	
VSEARCH version 2.17.1	Rognes et al., 2014	https://github.com/torognes/vsearch.git
DIAMOND version 0.9.24	Buchfink et al., 2015	https://github.com/bbuchfink/diamond.git
MEGAHIT	Li et al., 2015	
Kraken2	Wood et al., 2019	
HDBScan	McInnes et al., 2017	
Cytoscape	Shannon et al., 2003	
Other		